Your Name

Embedded Systems & ECE | IoT · Instrumentation & Control

Your City, Country | +1 555 000 0000 | you@example.com
linkedin.com/in/yourname | github.com/yourname | yourname.dev

Education

Example University Expected May 2027
B.S. in Electrical and Computer Engineering | GPA: 3.8/4.0 *Your City, Country*
- **Relevant Coursework:** Embedded Systems, Digital Logic, Control Systems, Signals, Microcontrollers.

Experience

Embedded Systems Intern Jun 2026 – Aug 2026
Example Devices Lab *Your City, Country*
- Built STM32 C firmware to poll 5 Modbus sensors over RS-485 at 115200 baud.
- Cut SPI telemetry latency 30% with DMA transfers, eliminating packet loss in 2-hour bench runs.
- Documented 12 fault cases and added recovery checks, reducing repeat lab debugging by 4 hours weekly.

Instrumentation Intern Jan 2026 – May 2026
Acme Controls Group *Your City, Country*
- Tuned an ESP32 PID loop to hold a test chamber within ±1 °C from 20–60 °C.
- Calibrated 3 RTDs against a reference probe, lowering average temperature error from 2.1 to 0.8 °C.

Firmware Intern May 2025 – Aug 2025
Example Sensor Works *Remote*
- Added nRF52 sleep scheduling to a BLE sensor, extending estimated battery life from 8 to 14 months.
- Traced intermittent I2C lockups with a logic analyzer and fixed the bus recovery sequence.

Projects

Balancing Robot | *C++, FreeRTOS, STM32* | GitHub Sep 2025 – Dec 2025
- Separated IMU sampling, control, and logging into 3 FreeRTOS tasks at a 100 Hz loop rate.
- Stabilized a two-wheel prototype with a tuned PID controller, recovering from 10° tilt tests.

LoRa Sensor Node | *C, Zephyr RTOS, LoRaWAN* | GitHub Feb 2025 – May 2025
- Sent pressure readings over LoRaWAN at 1 km across campus with acknowledgements in 9 of 10 trials.
- Measured 22 uA sleep current on a bench supply after tuning sensor and radio duty cycles.

Leadership & Activities

Student Lab Mentor Sep 2025 – Present
Example University ECE Club *Your City, Country*
- Guided 8 first-year students through soldering and sensor labs, helping each complete a working demo.

Robotics Club Member Sep 2024 – May 2025
Example University *Your City, Country*
- Assembled a line-following robot for a 6-team campus challenge and finished in the top 3.

Skills

Languages & Systems: C, C++, Python, ARM Cortex-M, FreeRTOS, Zephyr, Linux
Protocols & Buses: UART, SPI, I2C, CAN, Modbus RTU, RS-485, BLE, LoRaWAN
Hardware & Tools: KiCad, oscilloscope, logic analyzer, JTAG/SWD, soldering, Git
Instrumentation & Control: PID tuning, RTDs, 4–20 mA loops, MATLAB/Simulink

Certifications

Example Embedded C Course Certificate; Example University PCB Design Workshop